

Ranking Transfers, Thresholds Don’t: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show that a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus even when its decision *threshold* does not, and build an unsupervised *test-time adaptation* (TTA) method on that asymmetry: self-train on confident-ranked pseudo-labels from the extreme quantiles of unlabeled target scores, enforce channel-consistency, and adapt only LayerNorm and the classifier head—no target labels, no source data. On a four-target, ten-seed leave-one-corpus-out protocol, EER improves on three targets (In-the-Wild 12.78 $\rightarrow$ 11.33%, Arabic 22.50 $\rightarrow$ 20.99%, ASvspoof2019 5.03 $\rightarrow$ 3.47%; two survive Holm correction) and is neutral on the fourth, the one with the weakest source ranking. Applied unmodified to nine *publicly released* checkpoints we did not train—five architectures, one a spectrogram transformer—over 257 cross-corpus cells, it closes the gap between accuracy at the shipped threshold and the accuracy the ranking already supports (7.8 $\rightarrow$ 1.0 points, $p \approx 10^{-20}$, adapting 16,706 parameters) while its effect on ranking is not significant: the mechanism is threshold repair. Consistent with this, a label-free rule that moves the threshold to the median score is statistically indistinguishable from the full method over 133 cells (both ~ 1 point below a labelled oracle), and both break together when the target pool is skewed from class balance. The EER gains that do occur track *calibration headroom*: on two corpora sharing no lineage with the source pool (WaveFake, 2024 commercial TTS) adaptation improves EER by +2.5 and +2.9 points ($p=0.002$), larger than on the original targets. Entropy-minimisation TTA (Tent) is instead unpredictably catastrophic, and a non-SSL backbone never reaches a ranking worth adapting.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [12].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high ROC-AUC while its accuracy at a fixed threshold collapses—it still *ranks* clips correctly but its operating point is miscalibrated. Because ranking transfers, its most-confident pre-

dictions on unlabeled target audio are trustworthy enough to adapt on, *at test time*, without any labels.

Our contributions: (i) an operational use of the ranking—calibration asymmetry—ranking transfers cross-corpus even when the threshold does not—as the precondition that makes unsupervised self-training viable; (ii) a simple, stable TTA method—confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head; (iii) a four-target, multilingual, ten-seed leave-one-corpus-out evaluation against seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline), including the one target where the method does not help; (iv) a mechanistic analysis—proxy $\mathcal{A}$ -distance, a label-free median-threshold control that matches the full method over 133 cells, and a class-balance skew sweep—placing the mechanism at *operating-point repair under a balanced-prior assumption*, not domain-invariance or improved separability; and (v) evidence that the method acts as a *label-free wrapper for third-party detectors*, restoring the operating point of nine publicly released checkpoints across five architectures and ten corpora (257 cells) without retraining them.

Related work. Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]; we additionally test whether the cross-corpus gap is an SSL question at all via a from-scratch, non-SSL backbone. Train-time domain generalisation (DG) methods align real and separate fake across source domains (aggregation-separation [5]) or run gradient-reversal domain-adversarial training over source and target jointly (DANN) [8]; both act at training time rather than at deployment. Entropy minimisation (Tent) [7] is the standard TTA recipe in vision but, as we show, is unstable here; batch-statistics recalibration [9] is a no-op for LayerNorm architectures. We borrow the proxy $\mathcal{A}$ -distance from Ben-David et al. [10] to test mechanism.

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake=1). A detector f_θ pools frame features into an embedding e and outputs $s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y=1 \mid x)$: XLS-R-300M with only its top four transformer layers trainable,

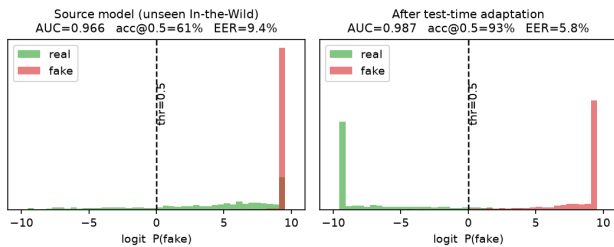


Fig. 1. Detector scores on unseen In-the-Wild audio. **Left:** the source model pushes many real clips past $\tau=0.5$ (accuracy well below AUC) yet still ranks most real below fake. **Right:** TTA shifts the real mass back across the threshold.

attentive temporal pooling, and a linear head. Its normalisation is *LayerNorm*, which carries no running batch statistics, so a batch-statistics-recalibration baseline is architecturally a no-op here and confidence-driven gradient adaptation is the minimal mechanism that can move these layers at all.

The ranking-calibration gap. Ranking quality $AUC = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free and invariant to reparametrisation of s_θ ; accuracy at a fixed threshold τ is not. Empirically, AUC degrades far less than accuracy at $\tau=0.5$ under source $\rightarrow$ target shift (Fig. 1). We therefore trust score *order*: the extreme quantiles of unlabeled target scores are higher-precision pseudo-labels than a fixed threshold, even when s_θ is mis-calibrated. The fixed-quantile split Q_q vs. Q_{1-q} implicitly assumes both a reliable source ranking and roughly balanced target classes; we scope the method to that regime, and Sec. IV and Sec. V show what happens when each assumption breaks.

Test-time adaptation. Each epoch, we assign confident pseudo-labels from the q - and $(1-q)$ -quantiles of target scores, ignoring the middle; self-train with cross-entropy on this confident set ($\mathcal{L}_{ST}$); and enforce prediction invariance under a stochastic channel perturbation $a(\cdot)$ (gain+noise) via an MSE consistency term $\mathcal{L}_{cons}$ between clean and perturbed views (stop-gradient on the clean target). The objective is

$$\min_{\{\gamma, \beta\}, \phi} \mathcal{L}_{ST} + \lambda \mathcal{L}_{cons}, \quad \lambda=0.3, q=0.3, E=4, \quad (1)$$

optimising *only* the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned layers and the head ϕ ; the rest stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse to which entropy minimisation is prone.

Domain divergence. To test whether adaptation works by making source and target indistinguishable in embedding space (as adversarial DG methods explicitly optimise for), we compute a proxy $\mathcal{A}$ -distance $\hat{d}_A = 2(1 - 2\epsilon_h)$ following Ben-David et al. [10], where ϵ_h is a linear domain classifier’s held-out error on frozen embeddings, before and after adaptation.

III. EXPERIMENTAL SETUP

Protocol. Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [13]; D2 (LibriSpeech bona fide speech vs. fakes from eight open TTS/voice-cloning systems); In-the-Wild [12]; and the Arabic Audio Deepfake

TABLE I
CROSS-CORPUS EER % / AUC, MEAN OVER 10 SEEDS
(LEAVE-ONE-CORPUS-OUT, TRANSDUCTIVE). BEST OF {SOURCE, TENT, ST-ONLY, OURS} PER COLUMN IN **BOLD**.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
Tent [7]	23.47 $\pm$ 21.96 / .776	35.44 $\pm$ 3.20 / .674	27.34 $\pm$ 11.21 / .735	47.35 $\pm$ 5.03 / .527
Self-training only	4.20 / .987	33.39 / .709	11.55 / .950	22.19 / .846
Ours (TTA)	3.47 / .994	33.74 / .727	11.33 / .958	20.99 / .866
Oracle (target-supervised)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

set (ArAD) [15]. For each held-out target the source pool merges the other three corpora with a 38-language MLAAD subset [14] (fake-only); held-out target audio, including its MLAAD-overlapping languages, is never in any source pool, and six MLAAD languages are held out of every pool to test unseen-language generalisation.

Comparison points. Source-only; Tent [7] (entropy minimisation, same trainable set); BN-only (AdaBN [9], no gradient step); self-training-only ($\lambda=0$); Oracle (target-supervised upper bound); RawNet2Lite (from-scratch non-SSL backbone [4], trained identically); DANN [8], trained to convergence with the held-out target’s *unlabeled* inputs through a gradient-reversal branch—more privileged access than ours; and an ASDG-style aggregation-separation baseline [5], which aligns across *source* domains only.

Implementation. 16 kHz audio, 3 s crops, top-4-layer XLS-R fine-tuning, Adam $lr 2 \times 10^{-4}$, batch 32, source trained 8 epochs. Adaptation: same batch and crop, $lr 1 \times 10^{-4}$, $E=4$ passes; $a(\cdot)$ is per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise ($\sigma=0.005$); the head carries over from the source model. Tables II and I and every significance test use ten seeds per target for every method; the third-party, $\mathcal{A}$ -distance and 2021-DF studies give their run counts inline. Code and manifests: [URL withheld].

IV. RESULTS

Table I is the main result. Our method wins on three of four targets: ASVspoof2019 (5.03 $\rightarrow$ 3.47%, 10/10 seeds), Arabic (22.50 $\rightarrow$ 20.99%, 10/10), In-the-Wild (12.78 $\rightarrow$ 11.33%, 8/10); on LibriSpeech-TTS it is EER-neutral (33.54 $\rightarrow$ 33.74%, improved in only 5/10 seeds), and that is also the target with the lowest source AUC (0.73 vs. ≥ 0.84 elsewhere). It is tempting to read source AUC as the precondition, but the apparent $r=+0.86$ is an artefact of four clusters: within each target the sign is reversed or absent ($r=-0.77$, -0.64 , -0.64 , $+0.12$; Fig. 2), and the third-party breadth arm below kills it outright. The precondition that does hold—calibration headroom, and class balance—is established there.

Significance. Ten seeds per target make a *per-target* Wilcoxon possible at all ($n=10$ floor 0.002). Against source-only, ours is significant on ASVspoof2019 and Arabic ($p=0.0020$) and In-the-Wild ($p=0.0293$), not LibriSpeech-TTS ($p=0.23$); Holm correction keeps the first two ($p_{adj}=0.0078$) and makes In-the-Wild nominal (0.0586), with a bootstrap agreeing. Pooled over all 40 pairs (descriptive only, clustered) ours beats source-only ($p=2.4 \times 10^{-5}$), Tent

(1.5×10^{-7}), self-training-only (0.0056) and both DG baselines (30/40 each).

Tent is not merely unstable, it is unpredictably catastrophic. On ASVspoof2019 its per-seed EER splits five seeds near source-only and four at or near chance with no warning sign (std 21.96, mean 23.47); on Arabic it is a *consistent* chance-level failure ($47.35 \pm 5.03\%$, 9/10 seeds). Our method shows no comparable collapse on any target (largest std 2.01). The failure is a ranking failure first: run unmodified on the strongest and weakest public checkpoints below, Tent collapses to chance on one ($16.41 \rightarrow 49.80\%$ EER, AUC $.918 \rightarrow .502$) [17] but only degrades AUC on the other ($.986 \rightarrow .979$) [16]—an EER collapse appears only where there was AUC headroom to lose.

Transfer to third-party detectors. Every result above adapts our own source model, so we apply the published configuration unmodified to nine released detectors we did not train—XLS-R+AASIST checkpoints trained on WaveFake [16] and on ASVspoof2019 [17], wav2vec2 sequence classifiers, and an Audio Spectrogram Transformer [11]—across ten corpora, 257 cells in all.

The consistent effect is *re-calibration*. Writing $\delta = (100 - \text{EER}) - \text{acc}@0.5$ for the accuracy a model’s ranking supports but its shipped threshold does not deliver, adaptation drives δ from 7.77 to 1.04 points (187/257 cells, Wilcoxon $p = 8.1 \times 10^{-20}$), and $\text{corr}(\delta_{\text{before}}, \delta_{\text{after}}) = +0.22$: where it lands barely depends on where it started. Within families it holds wherever there is a deficit to close—the spectrogram transformer 39/39 ($p = 5 \times 10^{-8}$), the WaveFake-trained XLS-R 39/40, one ASVspoof2019-trained XLS-R 38/40 ($p < 10^{-11}$)—is a no-op on families whose shipped threshold is already near-optimal, and *worsens* calibration on one wav2vec2 classifier (2/12). That the same LayerNorm-and-head selection rule, designed against XLS-R, re-calibrates a mel-spectrogram transformer unchanged is the clearest architecture-independence evidence here.

Ranking is a different matter. Pooled over the same cells the EER change is *not* significant ($+0.46\%$, $p = 0.16$; AUC $+0.005$, $p = 0.04$). Per cell at ten seeds with Holm correction, 8/12 tested cells are significant—but four of those are significant *degradations*, the largest $38.13 \rightarrow 41.91\%$. Our earlier single-corpus result (In-the-Wild, four checkpoints, all improved, $16.41 \rightarrow 10.10\%$ on the weakest) reproduces but does not generalise. It is not memorisation: adapting on half of In-the-Wild and scoring the disjoint half gives 4.08% vs. 4.06% transductive.

The method is a label-free threshold rule. If the entire benefit is a shifted operating point, gradient descent should not be needed to obtain it. For each source checkpoint we score the pool once and threshold at the *median* score (label-free; the split predicting 50% positive). Over 70 of our own cells the median rule reaches 75.3% accuracy vs. TTA’s 75.6% (labelled oracle 76.6%)—94% of TTA’s gain with no training and none of the 16,706 adapted parameters. Over 63 third-party cells it *beats* TTA, 75.6 vs. 73.7% ($p = 0.04$, within 0.4 of the oracle): where adaptation degrades a checkpoint’s ranking a threshold

move cannot, so it is the safer of the two. Pooled over all 133 cells the two are statistically indistinguishable, both ~ 1 point under the labelled oracle. Both assume a class-balanced pool and break together when that fails (Sec. V). The gradient method is doing, expensively, what a one-line threshold move does at least as well.

Where EER gains do occur, they track calibration headroom, and lineage predicts headroom. Extending our own model past the four original targets: three ASVspoof2021 tracks (LA/DF/PA) erase the pooled EER benefit ($+0.88 \rightarrow -0.68$, $n=70$, $p=0.80$)—but these are relatives of ASVspoof2019, which is *in* the source pool, so they arrive already well-thresholded (deficit -2.7 to $+3.2$) with nothing to repair. Two corpora that share no lineage with any source—WaveFake (vocoder artefacts over LJSpeech) [16] and a 2024 commercial-TTS set (ElevenLabs, Polly, Kokoro, Hume, Speechify)—instead arrive badly mis-thresholded (deficit 15.6, 5.8) and adaptation improves their EER by $+2.5$ and $+2.9$ points (10 seeds each, $p=0.002$), *larger* than any original target. Pooling the six genuinely independent targets gives $+1.48$ EER ($n=60$, $p < 10^{-8}$). The positive EER numbers are contingent not on which targets were to hand but on calibration headroom, which lineage-relatedness predicts.

The adaptation budget is a confound we introduced. E counts epochs, so the number of updates scales with pool size: $E=4$ over In-the-Wild’s 31,779 clips is $5.3 \times$ the steps of $E=4$ over a 6,000-clip pool. On our own model at ten seeds, $E=32$ lifts the Arabic gain from $+1.41$ to $+3.65$ EER on 10/10 seeds ($p=0.002$), while on LibriSpeech-TTS—the weakest source ranking—it does not help (2/10). Handed the same $8 \times$ budget the baselines do not benefit (Tent worsens by 5 EER points on Arabic, self-training-only is flat), so the gain is the method’s, not the compute’s. It also reverses three of the four significant third-party degradations ($-3.79 \rightarrow +6.35$ on the largest, at $E=32$); the exception is the chance-AUC cell. E belongs in step counts, not epochs; but more steps is not a free win, so we report the defect rather than re-tabulate around it.

Backbone and prior DG baselines. RawNet2Lite (no SSL pretraining) stays near chance on every target (AUC 0.40–0.65) even retrained for 40 epochs with stronger regularisation, so cross-corpus viability starts from SSL pretraining, not the TTA step. Both train-time DG baselines are beaten pooled over all 40 pairs (30/40 each); per target we beat DANN on two of four ($p \leq 0.0098$; tie on In-the-Wild and LibriSpeech-TTS) and ASDG on three of four, with ASDG better on LibriSpeech-TTS (32.64 vs. 33.74%). DANN sees the held-out target’s unlabeled audio every training step yet is worse, costs $3 \times$ as much, and is seed-unstable (ASVspoof2019 EER 3.23–14.85%); ASDG beats plain source-only on only two of four targets (Table II). Neither is a reliable route to cross-corpus robustness here.

Mechanism. An inductive check (adapt and evaluate on disjoint target halves) rules out transductive memorisation. The proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *increase* slightly after adaptation (Arabic $\hat{d}_{\mathcal{A}}: 1.545 \rightarrow 1.606$),

TABLE II

TEST-TIME ADAPTATION VS. TWO TRAIN-TIME DOMAIN-GENERALISATION BASELINES, MEAN EER % / AUC OVER THE SAME 10 SEEDS FOR EVERY METHOD. BEST PER ROW IN **BOLD**. OURS WINS 30/40 (TARGET, SEED) PAIRS AGAINST EACH; PER TARGET WE BEAT ASDG ON THREE OF FOUR AND DANN ON TWO OF FOUR, AND ASDG IS BETTER ON LIBRISPEECH-TTS. ONLY DANN SEES TARGET AUDIO DURING TRAINING. ASDG RAN ON DIFFERENT HARDWARE, SO ITS WALL-CLOCK IS NOT COMPARED.

Target	Source	DANN [8]	ASDG [5]	Ours
ASVspoof2019	5.03 / .989	7.35 / .966	4.98 / .987	3.47 / .994
LibriSpeech-TTS	33.54 / .727	34.43 / .717	32.64 / .749	33.74 / .727
In-the-Wild	12.78 / .945	13.34 / .924	17.33 / .891	11.33 / .958
Arabic	22.50 / .843	28.53 / .759	27.41 / .798	20.99 / .866

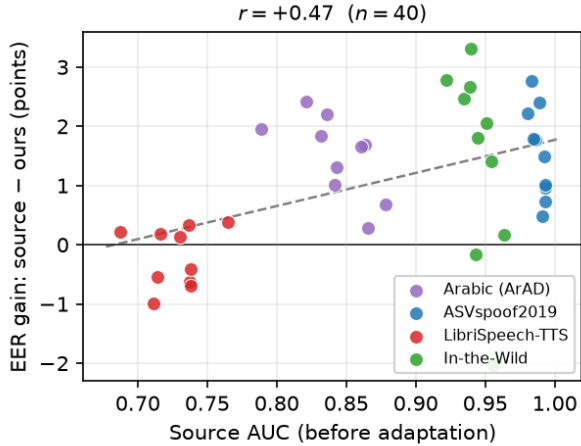


Fig. 2. EER gain (source—ours) vs. source AUC before adaptation, one point per (target, seed), all 40 runs. The association is between corpora, not within them: the four target means give $r=+0.86$ ($n=4$, $p=0.14$) and the pooled 40-point fit gives $r=+0.47$, but within each target the sign is reversed or absent ($r=-0.77$, -0.64 , -0.64 , $+0.12$). LibriSpeech-TTS (red, lowest AUC) is the only target with a negative mean gain.

the opposite of what domain-invariance predicts: adaptation re-shapes the boundary with rank-reliable pseudo-labels rather than erasing domain identity, consistent with DANN underperforming despite optimising that quantity. MLAAD fake-clip recall is uniform across seen and unseen languages (93.9 vs. 93.3%).

V. DISCUSSION AND LIMITATIONS

The method is transductive (unlabeled target inputs at adaptation time, the standard TTA setting); an inductive check reproduces every result including the LibriSpeech-TTS null. We report per-target, per-seed numbers throughout, so the limitations below can be read against the evidence:

- **Needs an already-reliable source ranking—and we cannot say how much of that is SSL.** A from-scratch or low-AUC starting point is not a regime we can rescue (Fig. 2). Our non-SSL backbone never reaches a ranking worth adapting (AUC 0.40–0.65), so it fails the precondition before TTA becomes applicable and we did

not run adaptation on it; every third-party checkpoint we adapt is itself a pretrained transformer.

- **Class balance is a second precondition, and it is the one the operator can check without labels.** The fixed-quantile rule labels the bottom q of the pool “confident real” whatever the true prior, so it targets a *balanced*-prior operating point. A controlled skew sweep (our model, four targets) confirms the mechanism: as the pool goes $50 \rightarrow 70 \rightarrow 90\%$ fake, the accuracy gain goes $+7.6 \rightarrow -3.5 \rightarrow -26.5$ points while ranking stays flat—at 90% fake, adaptation drags a 87.3% source model to 60.8%. This is the same failure seen once on the official ASVspoof2021-DF protocol (97.2%-spoof pool: source 26.33% EER, TTA 42.46%), and label-free corrections do not fix it (prevalence-weighted budget: 32.58% / 42.46% under two source recipes; q -curriculum: 42.50%). Deployment prevalence is normally known, so we state it as a precondition rather than estimate it from audio. (Our source model is also not state-of-the-art—8–9% EER is reported on 2021-DF by SSL+AASIST systems against our 26.33% source-only—so this is a *relative*-gain claim on a fixed backbone, to be re-verified on a stronger one.)
- **Fixed hyperparameters.** q and λ are label-free and not per-target optimal; a label-free λ schedule pays nothing at our scale (within 0.24 EER of a per-point oracle). E is not a hyperparameter but a mis-specified budget, as above.
- **Scale.** The main table is four targets, ten seeds; going from three to ten seeds made the per-target tests possible but shrank our DG margin to two per-target wins against DANN and three against ASDG. Breadth comes from the third-party arm (nine detectors, ten corpora, 257 cells) and six additional ten-seed targets for our own model, which is where the source-AUC ordering was falsified and the calibration account confirmed. The mechanistic claims (threshold rule, balanced-prior assumption) are established on our model and the label-free control; the two large HF checkpoints run at batch 8 not 16, and three of the nine third-party checkpoints have undocumented training data and are kept in a separate arm.

VI. CONCLUSION

Cross-corpus audio deepfake detection fails because the operating point is mis-calibrated, not because ranking is lost. Our unsupervised TTA repairs the operating point—closing the shipped-vs-attainable accuracy gap on nine third-party detectors over 257 cells ($p \approx 10^{-20}$) while leaving ranking unchanged—and a label-free move of the threshold to the median score matches it over 133 cells with no training; both require a class-balanced target pool. The EER gains that remain are conditional on calibration headroom, which lineage-relatedness predicts (independent WaveFake and commercial-TTS targets gain $+2.5/+2.9$ EER). Source AUC does not predict the gain and is retired. It is a re-calibration wrapper, not a detector.

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